3.20 Krull Dimension

Definition 3.20.1 (Krull Dimension)
Let A be a commutative ring. We say that a chain of distinct prime ideals

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n$$

*has **length** n.*

a) *The **Krull dimension** $\dim A$ of S is the maximum length of all chains of prime ideals.*
b) *The **height** of a prime ideal $\mathfrak{p}$, denoted $\operatorname{ht}(\mathfrak{p})$, is the maximum length of chains of prime ideals contained in $\mathfrak{p}$. More generally define $\operatorname{ht}(\mathfrak{a}) = \inf\{\operatorname{ht}(\mathfrak{p}) \mid \mathfrak{a} \subseteq \mathfrak{p}\}$. By (3.3.44) we may take the infimum over only minimal prime ideals.*
c) *The **dimension** of an ideal $\mathfrak{a} \lhd A$, denoted $\dim \mathfrak{a}$, is the maximum length of chains of prime ideals containing $\mathfrak{a}$.*

*We say A is **finite-dimensional** if $\dim A < \infty$. Observe that $\mathfrak{a} \subseteq \mathfrak{p} \iff \sqrt{\mathfrak{a}} \subseteq \mathfrak{p}$ for any prime ideal $\mathfrak{p}$ so*

$$\dim \mathfrak{a} = \dim \sqrt{\mathfrak{a}}$$

$$\operatorname{ht}(\mathfrak{a}) = \operatorname{ht}\left(\sqrt{\mathfrak{a}}\right)$$

and we may, without loss of generality, consider only radical ideals.

Definition 3.20.2
Let A be a commutative ring. We say a chain of prime ideals is

- ***maximal** if it's not properly contained in any chain*
- ***saturated** if $\mathfrak{p}_i \subseteq \mathfrak{p} \subseteq \mathfrak{p}_{i+1} \implies \mathfrak{p} = \mathfrak{p}_i$ or $\mathfrak{p} = \mathfrak{p}_{i+1}$.*

*We say that A is **biequidimensional** if every maximal chain has the same length (equal to $\dim A$).*

*We say A is **quasi-biequidimensional** if $A/\mathfrak{p}$ is **biequidimensional** for every minimal prime $\mathfrak{p}$. Note that **equidimensional** + **quasi-biequidimensional** $\iff$ **biequidimensional**.*

*We say that A is **catenary** if the length of a saturated chain*

$$\mathfrak{p}_0 \subsetneq \cdots \subsetneq \mathfrak{p}_n$$

depends only on $\mathfrak{p}_0$ and $\mathfrak{p}_n$ and is equal to $\operatorname{ht}(\mathfrak{p}_n/\mathfrak{p}_0)$

*We say that A is **equidimensional** if every minimal prime ideal has the same dimension (equal to $\dim A$). Note an **irreducible** ring has only one minimal prime ideal so is trivially equidimensional.*

*We say that A is **equicodimensional** if every maximal ideal has the same height (equal to $\dim A$).*

In order to connect this to the lattice-theoretic notion of Krull Dimension (see the Krull Dimension section of Foundations) we prove the following result (provided we consider the lattice of radical ideals ordered by *reverse* inclusion).

Proposition 3.20.3
Let A be a ring then the lattice of radical ideals $\operatorname{Rad}(A)$ is distributive, that is we have equality

$$\mathfrak{r}_1 \cap \sqrt{\mathfrak{r}_2 + \mathfrak{r}_3} = \sqrt{\mathfrak{r}_1 \cap \mathfrak{r}_2 + \mathfrak{r}_1 \cap \mathfrak{r}_3}$$

*Furthermore the meet-prime radical ideals are precisely the prime ideals. Therefore the lattice of radical ideals of a finite-dimensional Noetherian ring, ordered by **reverse inclusion**, is a Krull Lattice.*

Proof. Clearly it's enough to show that LHS $\subseteq$ RHS. Suppose $x \in$ LHS then $x \in \mathfrak{r}_1$ and $x^n = a + b$ where $a \in \mathfrak{r}_2$ and $b \in \mathfrak{r}_3$. Then $x^{n+1} = ax + bx \in \mathfrak{r}_1 \cap \mathfrak{r}_2 + \mathfrak{r}_1 \cap \mathfrak{r}_3$ whence $x \in$ RHS.

We've shown that prime ideals are meet-prime (3.3.40). Suppose $\mathfrak{r}$ is a meet-prime radical ideal, and $fg \in \mathfrak{r}$. Then we claim that

$$\sqrt{\mathfrak{r} + (f)} \cap \sqrt{\mathfrak{r} + (g)} \subseteq \mathfrak{r}$$

For $x \in \text{LHS} \implies x^n \in \mathfrak{r} + (f)$ and $x^m \in \mathfrak{r} + (g) \implies x^{n+m} \in \mathfrak{r} \implies x \in \mathfrak{r}$. As $\mathfrak{r}$ is meet-prime then for example $\sqrt{\mathfrak{r} + (f)} \subseteq \mathfrak{r}$ and in particular $f \in \mathfrak{r}$. Therefore $\mathfrak{r}$ is also prime. ∎

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Proposition 3.20.4 (Simple properties)
The following properties of Krull dimension hold

a) $\dim A = \dim A/N(A)$
b) $\mathrm{ht}(\mathfrak{p}) = \dim A_{\mathfrak{p}}$
c) $\dim \mathfrak{a} = \dim A/\mathfrak{a}$
d) $\dim \mathfrak{a} = \dim \mathfrak{a}/\mathfrak{b}$ *for any ideal* $\mathfrak{b} \subseteq \mathfrak{a}$
e) $\dim A = \sup_{\mathfrak{p}} \dim A_{\mathfrak{p}}$
f) ***codimension inequality*** $\mathrm{ht}(\mathfrak{p}) \geq \mathrm{ht}(\mathfrak{p}/\mathfrak{q}) + \mathrm{ht}(\mathfrak{q})$
g) $\dim k = 0$ *for any field* k
h) A principal ideal domain A *which is not a field has dimension* 1

Proof. a) By (3.3.56) there is an order-isomorphism between prime ideals of A containing $N(A)$ and prime ideals of $A/N(A)$. However by (3.3.46) all prime ideals of A contain $N(A)$, so there is a bijection between chains of A and chains of $A/N(A)$, and the result follows.

b) This follows similarly from (3.5.42).

c) This follows similarly from (3.3.56).

d) $\dim \mathfrak{a} = \dim A/\mathfrak{a} = \dim(A/\mathfrak{a})/(\mathfrak{a}/\mathfrak{b}) = \dim \mathfrak{a}/\mathfrak{b}$

e) This follows from (2.5.6)

f) This follows from (2.5.6)

g) The only (prime) ideal is (0)

h) By (...) every prime ideal (besides (0)) is maximal so every chain has length at most 1. ∎

Lemma 3.20.5
Let A *be a UFD and* $\mathfrak{p} \lhd A$ *a non-zero prime ideal. Then it contains a non-zero principal prime ideal* (p).

In particular $\mathrm{ht}(\mathfrak{p}) = 1$ *if and only if it is principal.*

Proof. Choose $0 \neq f \in \mathfrak{p}$. Then by definition it has a factorization into primes, and at least one must be in $\mathfrak{p}$, say p. Then (p) is prime by (3.15.9). Therefore if $\mathrm{ht}(\mathfrak{p}) = 1$ then it is equal to (p).

Conversely if $\mathfrak{q} \subseteq (p)$ then $(q) \subseteq \mathfrak{q} \subseteq (p)$ for q prime, which implies $p \mid q$. As q is irreducible (...) then $(p) = (q) = \mathfrak{q}$. Therefore $\mathrm{ht}((p)) = 1$. ∎

Proposition 3.20.6
Let A *be a Noetherian ring and* $\mathfrak{a}$ *a proper ideal. Then there are only finitely many minimal primes containing* $\mathfrak{a}$*, say* $\mathfrak{p}_1, ..., \mathfrak{p}_n$*. Further we have a decomposition*

$$\sqrt{\mathfrak{a}} = \bigcap_{i=1}^{n} \mathfrak{p}_i$$

which is irredundant (and the only such decomposition). Furthermore

$$\mathrm{ht}(\mathfrak{a}) = \min_i \mathrm{ht}(\mathfrak{p}_i)$$

$$\dim(\mathfrak{a}) = \max_i \dim(\mathfrak{p}_i)$$

Proof. This follows from (2.4.7) and (3.20.3) applied to the radical ideal $\sqrt{a}$. ∎

Remark 3.20.7
This is essentially the proof of [Kap74] (Theorem 87, 88).

We've noted in general (3.20.4) that the so-called codimension formula does not hold. However it holds in the following case, for essentially trivial reasons.

Proposition 3.20.8 (Codimension 1 formula)
Let A *be an irreducible Noetherian ring of finite Krull Dimension, and* $\mathfrak{a}$ *such that* $\dim(\mathfrak{a}) = \dim(A) - 1$ *then* $\mathrm{ht}(\mathfrak{a}) = 1$.

Proof. Apply (2.5.7). ∎

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In practice most rings of geometric interest are catenary or quasi-biequidimensional. We recall some properties and equivalent criteria for these cases.

Proposition 3.20.9 (Catenary Criteria)
Let A be a ring. Then the following are equivalent

a) A is catenary
b) For all prime ideals $\mathfrak{r} \subseteq \mathfrak{q} \subseteq \mathfrak{p}$ the following holds

$$\operatorname{ht}(\mathfrak{p}/\mathfrak{r}) = \operatorname{ht}(\mathfrak{p}/\mathfrak{q}) + \operatorname{ht}(\mathfrak{q}/\mathfrak{r})$$

When A is irreducible this is equivalent to the following condition

$$\operatorname{ht}(\mathfrak{p}) = \operatorname{ht}(\mathfrak{q}) + \operatorname{ht}(\mathfrak{p}/\mathfrak{q})$$

Proof. This is restatement of (2.5.9). ∎

Proposition 3.20.10 (Biequidimensional Criteria)
Let A be a ring. Then the following are equivalent

a) A is quasi-biequidimensional
b) A is catenary and $A/\mathfrak{p}$ is equicodimensional for every minimal prime $\mathfrak{p}$
c) A satisfies the formula

$$\dim \mathfrak{q} = \dim \mathfrak{p} + \operatorname{ht}(\mathfrak{p}/\mathfrak{q}) \quad \forall \mathfrak{q} \subseteq \mathfrak{p}$$

d) A satisfies c) whenever $\operatorname{ht}(\mathfrak{p}/\mathfrak{q}) = 1$ (i.e. $\mathfrak{q} \subsetneq \mathfrak{p}$ is saturated)

Furthermore

$$\operatorname{ht}(\mathfrak{p}) = \operatorname{ht}(\mathfrak{p}/\mathfrak{q}) + \operatorname{ht}(\mathfrak{q})$$

Proof. This is a restatement of (2.5.11). ∎

Proposition 3.20.11 (Irreducible Biequidimensional Criteria)
When A is irreducible the following are equivalent

a) A is biequidimensional
b) A is quasi-biequidimensional
c) A is catenary and equicodimensional
d) A satisfies the formula

$$\dim \mathfrak{q} = \dim \mathfrak{p} + \operatorname{ht}(\mathfrak{p}/\mathfrak{q}) \quad \forall \mathfrak{q} \subseteq \mathfrak{p}$$

e) A satisfies d) whenever $\operatorname{ht}(\mathfrak{p}/\mathfrak{q}) = 1$ (i.e. $\mathfrak{q} \subsetneq \mathfrak{p}$ is saturated)

Proof. This is a restatement of (2.5.14). ∎

Proposition 3.20.12 (Codimension Formula)
Let A be a quasi-biequidimensional ring, $\mathfrak{a}$ an ideal and $\mathfrak{p} \subset \mathfrak{a}$ a prime ideal. Then the following properties hold

$$\operatorname{ht}(\mathfrak{a}) = \operatorname{ht}(\mathfrak{a}/\mathfrak{p}) + \operatorname{ht}(\mathfrak{p})$$
$$\dim \mathfrak{p} = \dim \mathfrak{a} + \operatorname{ht}(\mathfrak{a}/\mathfrak{p})$$

When A is biequidimensional (e.g. quasi-biequidimensional and integral), then for all ideals $\mathfrak{a}$ we have the relation

$$\dim A = \dim \mathfrak{a} + \operatorname{ht}(\mathfrak{a})$$

Proof. This is a restatement of (2.5.13). ∎

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